

Supply Chain Analytics: Phase 2 – Logistics Performance & Supplier Reliability

Project Overview

Phase 2 transforms our "clean" data into actionable Key Performance Indicators (KPIs). By focusing on the temporal aspect of the supply chain—specifically the delta between **Expected Delivery** and **Actual Arrival**—we establish a scoring system for supplier and carrier reliability.

1. Core Analytical Foundation

We developed a centralized logic layer, `vw_shipment_performance_base`, which serves as the "engine" for all performance metrics.

Calculated Metrics:

- **Delay Months:** Calculated using `DATEDIFF` divided by 30.0 to normalize lead-time variance into a standard monthly scale.
 - **Binary Latency (is_late):** A boolean flag identifying any shipment that exceeded its `expected_delivery` date.
 - **Data Integrity Filter:** This view automatically excludes shipments with invalid date logic (arrival before shipment) or those linked to canceled orders, ensuring KPIs are not skewed by "bad data."
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2. Logistics & Warehouse Insights

We aggregated the performance data to identify systemic weaknesses in the distribution network.

Performance Distribution

- **On-Time vs. Late:** A high-level view of the fulfillment success rate across the entire network.
- **Delay Severity:** Categorizes late shipments into **Minor**, **Moderate**, or **Severe** (3+ months) buckets. This allows management to ignore "noise" and focus on catastrophic delays.

Infrastructure & Carrier Audit

- **vw_carrier_reliability:** Benchmarks carriers (e.g., UPS, FedEx, DHL) against each other using their specific `late_rate_pct` and `avg_delay_months`.

- **exposed_warehouses:** Identifies specific physical locations that are disproportionately affected by late arrivals, indicating potential regional logistics bottlenecks.

3. Supplier Risk Profiling

The centerpiece of this phase is the vw_supplier_delay_risk_profile. We moved beyond simple "On-Time" percentages to create a **weighted risk tiering system**.

Risk Tier Logic:

Tier	Criteria	Business Implication
High	Late Rate >= 40% AND Avg Delay => 3 Months	Critical Failure: High probability of long-term stockouts.
Medium	Late Rate =< 20%	Warning: Potential for inventory instability.
Low	Late Rate < 20% or No Late Shipments	Reliable: Strategic partners for high-volume SKUs.

4. Temporal Trend Analysis

Through delay_severity_by_month, we implemented time-series tracking. This allows the business to see if supplier performance is improving or degrading over time, helping to identify seasonal disruptions or the impact of global supply chain events.

5. Preliminary Findings

- **(On-Time Delivery):** 76% late rate was discovered, revealing incompetent shipping
- **Average Severe Delay:** average severe delay period got up to 10 months
- **Highest Risk Supplier:** 9 worst performing suppliers were identified for corrective actions
- **Weakest Carrier Link:** [LocalCarrier] proved to have the highest late rate percentage
- **Lead time analysis:** Delay severity became more prominent over time revealing worsening shipment performance

6. Next Steps

With performance metrics established, the project moves to **Phase 3: Inventory Risk Modeling**, where we will combine these supplier risk scores with live inventory levels to calculate a **Composite Risk Score (0-5)** for every product in the warehouse.